

MAT 426, Class 23: Inner Product Spaces

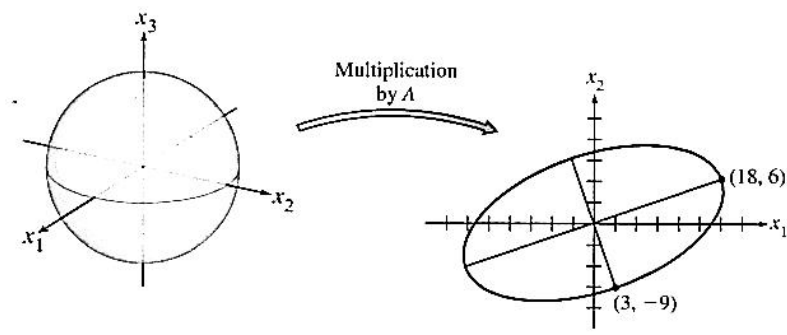
THEOREM *Let A be an $m \times n$ matrix with rank r . Then the image of the unit sphere in $\mathbb{R}^n$ under the matrix transformation that maps $\mathbf{x}$ to $A\mathbf{x}$ is*

(a) *the surface of an ellipsoid in $\mathbb{R}^m$ if $r = n$*

(b) *a solid ellipsoid in $\mathbb{R}^m$ if $r < n$*

Proof.

Example 1. Let $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$, Show that the linear transformation $T(\mathbf{x}) = A\mathbf{x}$ maps the unit sphere $\{\mathbf{x} : \|\mathbf{x}\| = 1\}$ onto the ellipse shown below.



Definition. An **inner product** on a vector space V is an operation that assigned every _____ of vectors $\mathbf{u}$ and $\mathbf{v}$ a _____ $\langle \mathbf{u}, \mathbf{v} \rangle$ such that the following properties hold for all vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in V and all scalars c :

- (1) $\langle \mathbf{u}, \mathbf{v} \rangle =$
- (2) $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle =$
- (3) $\langle c\mathbf{u}, \mathbf{v} \rangle =$
- (4) $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$ and $\langle \mathbf{u}, \mathbf{u} \rangle = 0$ if and only if _____

A vector space with an _____ is called an **inner product space**.

Example 2. Let $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ be vectors in $\mathbb{R}^2$. Show that $\langle \mathbf{u}, \mathbf{v} \rangle = 2u_1v_1 + 3u_2v_2$ defines an inner product.

Example 3. Let A be a symmetric, positive definite $n \times n$ matrix and let $\mathbf{u}$ and $\mathbf{v}$ be vectors in $\mathbb{R}^n$. Show that $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T A \mathbf{v}$ defines an inner product.

Example 4. Let f and g be in $\mathcal{C}[a, b]$, the vector space of all continuous functions on the closed interval $[a, b]$. Show that

$$\langle f, g \rangle = \int_a^b f(x)g(x) dx$$

defines an inner product on $[a, b]$.

Definition. Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in an inner product space V .

- (1) The **length** (or **norm**) of $\mathbf{v}$ is $\|\mathbf{v}\| =$ _____.
- (2) The **distance** between $\mathbf{u}$ and $\mathbf{v}$ is $d(\mathbf{u}, \mathbf{v}) =$ _____.
- (3) $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** if _____.

Example 5. Consider the inner product on $\mathcal{C}[0, 1]$, $\langle f, g \rangle = \int_0^1 f(x)g(x) dx$, from Example 3. Let $f(x) = x$ and $g(x) = 3x - 2$. Compute the following:

- (a) $\|f\|$
- (b) $d(f, g)$
- (c) $\langle f, g \rangle$

Example 6. Consider the inner product $\langle \mathbf{u}, \mathbf{v} \rangle = 2u_1v_1 + 3u_2v_2$ from Example 1. Sketch the “unit circle” corresponding to this inner product.

Definition.

- An **orthogonal set** of vectors in an inner product space V is a set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ of vectors from V such that _____ whenever _____.
- An **orthonormal set** of vectors is an _____ set of _____ vectors.
- An **orthogonal basis** for a subspace W of V is a _____ that is an _____ set. An **orthonormal basis** for a subspace W of V is a _____ that is an _____ set.

Example 7. Construct an orthogonal basis for $\mathcal{P}_2$ (the set of all polynomials of degree at most 2) with respect to the inner product

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx$$

by applying the Gram-Schmidt process to the basis $\{1, x, x^2\}$.

Definition. Suppose that W is a subspace of a vector space V that has an orthogonal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$. The **orthogonal projection** $\text{proj}_W(\mathbf{v})$ of a vector $\mathbf{v}$ onto the subspace W is

$$\text{proj}_W(\mathbf{v}) =$$

The **component of $\mathbf{v}$ orthogonal to W** is the vector

$$\text{perp}_W(\mathbf{v}) =$$

CAUCHY-SCHWARZ INEQUALITY Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in an inner product space V . Then,

$$| \langle \mathbf{u}, \mathbf{v} \rangle | \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

with equality holding if and only if $\mathbf{u}$ and $\mathbf{v}$ are scalar multiples of each other.

Proof.